CS315 · ALGORITHMS ANALYSIS & DESIGN

Midterm 461 Solutions

Midterm 461 · 25 marks

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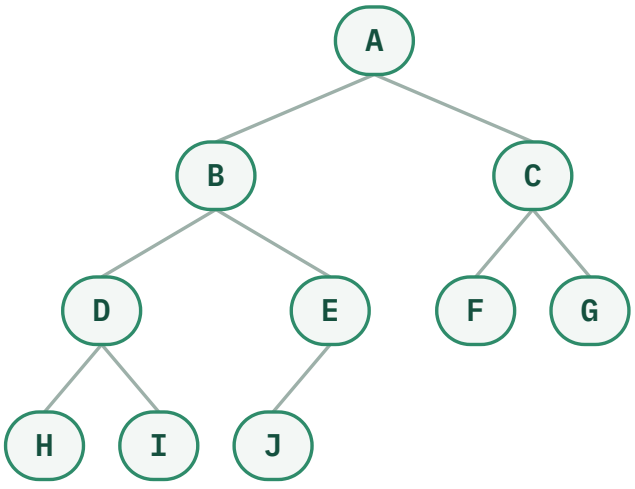
01 Solution Q1: True / False

#	Answer	Reason
1	**F**	The definition is a step-by-step procedure in a finite amount of time. The statement reverses it.
2	**T**	The words of the Lecture 1 slide.
3	**F**	The slide says "in time about n^2".
4	**T**	The words of the Lecture 3 slide.
5	**F** *	The items must first be sorted by value ÷ weight, and sorting takes `O(n log n)`.
6	**F**	It is the other way round: there are many feasible solutions, and the optimal one is the best of them.

* Item 5 depends on the greedy lecture's slide, which I have not received. The textbook answer is `O(n log n)`. Check the slide.

02 Solution Q2: MCQ

#	Answer	Reason
1	**(F,I,N,A) and (L,E,X,A,M)**	9 elements. The pseudocode takes the first $\frac{n}{2}$ elements, which is 4, and the remaining 5.
2	**Insertions are frequent and searches are rare**	Insertion is `O(1)` and search is `O(n)`.
3	**Characterize the structure of an optimal solution**	The first of the four DP steps.
4	**4**	The LCS is `GTAB`. The table is below.
5	**O(n log n)**	`n - 1` merges, each needing priority-queue operations of `O(log n)`.
6	**010**	The path from A to J: left (0) to B, right (1) to E, left (0) to J.



The LCS table for item 4:

	∅	G	X	T	X	A	Y	B
∅	0	0	0	0	0	0	0	0
A	0	0	0	0	0	1	1	1
G	0	1	1	1	1	1	1	1
G	0	1	1	1	1	1	1	1
T	0	1	1	2	2	2	2	2
A	0	1	1	2	2	3	3	3
B	0	1	1	2	2	3	3	4

The four DP steps in order: characterize the structure, recursively define the value, compute the value, then construct an optimal solution.

03 Solution Q3: double hashing

```
probe j:   (h(k) + j·d(k)) mod 10        j = 0, 1, 2, …
```

Key	k mod 10	k mod 9	Probes	Cell
19	9	1	9	9
16	6	7	6	6
90	0	0	0	0
18	8	0	8	8
60	0	6	0 → 6 → 2	2
89	9	8	9 → 7	7
70	0	7	0 → 7 → 4	4

0	1	2	3	4	5	6	7	8	9
90		60		70		16	89	18	19

```
60:  h = 0 taken,  d = 60 mod 9 = 6   →   (0 + 6) mod 10 = 6 taken   →
(0 + 12) mod 10 = 2
```

```
89:  h = 9 taken,  d = 89 mod 9 = 8   →   (9 + 8) mod 10 = 7
```

```
70:  h = 0 taken,  d = 70 mod 9 = 7   →   (0 + 7) mod 10 = 7 taken   →
(0 + 14) mod 10 = 4
```

The student's calculations visible in the margin of the paper match these steps.

Note: `d(90) = 0` and `d(18) = 0`, which breaks the lecture's rule that `d(k)` cannot be zero. It has no effect here because 90 and 18 were placed on their first probe.

04 Solution Q4: matrix chain

Multiplying a `p × q` matrix by a `q × r` matrix costs `p·q·r` operations.

```
AB  =  5 × 10 × 3   =  150
```

```
BC  =  10 × 3 × 12  =  360
```

```
ABC:   (AB)C  =  150 + 0 + 5 × 3 × 12   =  330     ← min
```

```
       A(BC)  =  0 + 360 + 5 × 10 × 12  =  960
```

	A	B	C
A	0	150	**330**
B	–	0	360
C	–	–	0

```
Minimum operations = 330
```

```
Best order of multiplication:   ((AB)C)
```

05 Solution Q5: fractional knapsack

v	40	30	20
w	20	10	5
v ÷ w	2	3	4

We take in order of `v ÷ w`, largest first:

```
Item 3:   all 5        →   value 20      remaining capacity 15
```

```
Item 2:   all 10       →   value 30      remaining capacity 5
```

```
Item 1:   5 out of 20  →   value 5 × 2 = 10
```

```
The maximum value = 20 + 30 + 10 = 60
```